

Rockchip GMS测试 摄像头模块帮助文档

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前言

概述

本文提供了GMS测试过程中Camera模块常见fail项的处理方法。

产品版本

Android 14.0及以上版本

读者对象

本文档（本指南）主要适用于以下工程师：

技术支持工程师

软件开发工程师

修订记录

版本号	作者	修改日期	修改说明
V1.0.0	王潘祯撰	2025-04-18	初始版本

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1. 概述

GMS测试包中包含大量摄像头相关（Camera）的模块，测试过程中会出现各种各样的fail，一些fail项需要找代码开发者去解决，一些fail项可以通过修改配置自行解决。本文提供了测试模块快速测试方法以及常见fail项解决办法。

2. 命令行测试

GMS测试正常流程是需要机器连接Ubuntu系统，并通过GMS测试包完成测试，测试时间相对比较久，在debug过程中如果通过正常流程去验证修改，效率偏低。

可以通过adb shell用命令行的方式通过am instrument启动Instrumentation测试。

以下以CtsCameraTestCases为例详细说明。

2.1 测试环境准备

2.1.1 CTS测试应用安装及权限设置

apk路径：

```
android-cts兼容性测试包/android-cts/testcases/CtsCameraTestCases/arm64/CtsCameraTestCases.apk
```

apk安装：

```
adb install CtsCameraTestCases.apk
```

权限设置：

```
Settings--Apps--All apps--android.camera.cts  
手动打开所需权限
```

2.2 测试命令

以CtsCameraTestCases中某一个case为例单测说明。

某一单项case：

```
android.hardware.camera2.cts.CaptureRequestTest#testEffectModes[1]
```

测试完整命令：

```
am instrument -w -r -e debug false -e class
android.hardware.camera2.cts.CaptureRequestTest#testEffectModes[1]
android.camera.cts/androidx.test.runner.AndroidJUnitRunner
```

测试结果:

```
130|rk3576_tablet:/ #
am instrument -w -r -e debug false -e class
android.hardware.camera2.cts.CaptureRequestTest#testEffectModes[1]
android.camera.cts/androidx.test.runner.AndroidJUnitRunner
INSTRUMENTATION_STATUS: class=android.hardware.camera2.cts.CaptureRequestTest
INSTRUMENTATION_STATUS: current=1
INSTRUMENTATION_STATUS: id=AndroidJUnitRunner
INSTRUMENTATION_STATUS: numtests=1
INSTRUMENTATION_STATUS: stream=
android.hardware.camera2.cts.CaptureRequestTest:
INSTRUMENTATION_STATUS: test=testEffectModes[1]
INSTRUMENTATION_STATUS_CODE: 1
INSTRUMENTATION_STATUS: class=android.hardware.camera2.cts.CaptureRequestTest
INSTRUMENTATION_STATUS: current=1
INSTRUMENTATION_STATUS: id=AndroidJUnitRunner
INSTRUMENTATION_STATUS: numtests=1
INSTRUMENTATION_STATUS: stream=.
INSTRUMENTATION_STATUS: test=testEffectModes[1]
INSTRUMENTATION_STATUS_CODE: 0
INSTRUMENTATION_RESULT: stream=

Time: 5.702

OK (1 test)

INSTRUMENTATION_CODE: -1
```

2.2.1 命令行说明

2.2.1.1 命令行格式

```
am instrument -w -r -e class testcase test_package/runner_class
```

2.2.1.2 参数说明

- w: 保持adb shell打开直至测试完成
- r: 以原始形式输出测试结果
- e <NAME> <VALUE>: set argument <NAME> to <VALUE>. 提供过滤器和参数

2.2.1.3 获取test_package/runner_class

通过pm list instrumentation找test_package/runner_class

```
rk3576_tablet:/ # pm list instrumentation
instrumentation:android.camera.cts/androidx.test.runner.AndroidJUnitRunner
(target=android.camera.cts)
```

CtsCameraTestCases的test_package/runner_class是：
android.camera.cts/androidx.test.runner.AndroidJUnitRunner

2.2.1.4 testcase

- 整个模块

testcase: CtsCameraTestCases

- 模块中的一个大类集合

testcase: android.hardware.camera2.cts.CaptureRequestTest

测试命令：

```
am instrument -w -r -e debug false -e class
android.hardware.camera2.cts.CaptureRequestTest
android.camera.cts/androidx.test.runner.AndroidJUnitRunner
```

- 模块中的一个大类集合中的某一个case

testcase: android.hardware.camera2.cts.CaptureRequestTest#testEffectModes[1]

测试命令：

```
am instrument -w -r -e debug false -e class
android.hardware.camera2.cts.CaptureRequestTest#testEffectModes[1]
android.camera.cts/androidx.test.runner.AndroidJUnitRunner
```

3. Camera feature

3.1 Camera feature 配置情况

```
rk3576_tablet:/ # pm list features | grep cam
feature:android.hardware.camera //后置Camera
feature:android.hardware.camera.any //包含camera feature默认会有
feature:android.hardware.camera.external //对应USB Camera
feature:android.hardware.camera.front //前置Camera
对应文件：
rk3576_tablet:/ # ls vendor/etc/permissions/android.hardware.camera.*
vendor/etc/permissions/android.hardware.camera.external.xml
vendor/etc/permissions/android.hardware.camera.front.xml
vendor/etc/permissions/android.hardware.camera.xml
```

3.2 Camera feature 选配位置

sdk/device/rockchip/rk35xx 目录的 BoardConfig.mk配置支持的camera类型

```
# camera enable
BOARD_CAMERA_SUPPORT := true //配置是否支持MIPI Camera
BOARD_CAMERA_SUPPORT_EXT := true //配置是否支持USB Camera
```

sdk/device/rockchip/common/modules\$ 中 camera_aidl.mk 将对应的feature文件编译进板子

```
ifeq ($(BOARD_CAMERA_SUPPORT_EXT),true)
PRODUCT_COPY_FILES += \
frameworks/native/data/etc/android.hardware.camera.external.xml:$(TARGET_COPY_OUT_VENDOR)/etc/permissions/android.hardware.camera.external.xml
endif

ifeq ($(BOARD_CAMERA_SUPPORT),true)
PRODUCT_COPY_FILES += \
frameworks/native/data/etc/android.hardware.camera.xml:$(TARGET_COPY_OUT_VENDOR)/etc/permissions/android.hardware.camera.xml \
frameworks/native/data/etc/android.hardware.camera.front.xml:$(TARGET_COPY_OUT_VENDOR)/etc/permissions/android.hardware.camera.front.xml
endif
```

4. Android SDK版本

Android发布的不同SDK版本，提交点差异，可能测试失败项都不一样；所以提交redmine时候，建议同步提供板子目前使用的sdk版本情况。

获取SDK版本命令：`getprop | grep rksdk`

示例如下：

```
rk3576_tablet:/ # getprop | grep rksdk
[ro.rksdk.version]: [ANDROID14_RKR6]
```

5. CTS

5.1 Android 14.0 USB Camera CTS 常见失败项处理办法

5.1.1 帧率相关

5.1.1.1 测试项

```
android.hardware.camera2.cts.CaptureRequestTest#testNoiseReductionModeControl[1]  
android.hardware.cts.CameraGLTest#testCameraToSurfaceTextureMetadata  
android.hardware.cts.CameraTest#testPreviewFpsRange
```

5.1.1.2 报错日志

- android.hardware.camera2.cts.CaptureRequestTest#testNoiseReductionModeControl[1]

```
java.lang.Exception: Test failed for camera 111: Frame duration must be in the  
range of [200000000, 500000000], value 110679333 is out of range [170000000,  
575000000]  
    at  
    android.hardware.camera2.cts.testcases.Camera2SurfaceViewTestCase.tearDown(Camera  
2SurfaceViewTestCase.java:176)  
    at  
    android.hardware.camera2.cts.CaptureRequestTest.tearDown(CaptureRequestTest.java:  
152)
```

- android.hardware.cts.CameraGLTest#testCameraToSurfaceTextureMetadata

```
java.lang.AssertionError: Too many frame intervals out of frame rate bounds: 98,  
limit 10  
    at org.junit.Assert.fail(Assert.java:89)  
    at org.junit.Assert.assertTrue(Assert.java:42)  
    at android.hardware.cts.CameraGLTest$6.run(CameraGLTest.java:665)  
    at android.hardware.cts.CameraGLTest.runForAllCameras(CameraGLTest.java:346)  
    at  
    android.hardware.cts.CameraGLTest.testCameraToSurfaceTextureMetadata(CameraGLTest  
.java:532)
```

- android.hardware.cts.CameraTest#testPreviewFpsRange

```
Test failed to run to completion. Reason: 'Test run failed to complete. Expected  
777 tests, received 755'. Check device logcat for details
```

5.1.1.3 解决办法

sdkhardware/rockchip/camera_aidl\$ 目录添加如下修改，如没有更新SDK或找FAE获取

```
commit 21be8bc6487b27800c5a25612980fc400d55e9f3  
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>  
Date:   Fri Aug 23 10:21:08 2024 +0800  
  
    device: usb camera: default use fpsranges [15:30] fix cts issue  
  
    Test: CtsCameraTestCases  
  
    android.hardware.camera2.cts.CaptureRequestTest#testNoiseReductionModeControl[1]
```

```
failed log:
Frame duration must be in the range of [500000000, 1000000000],
value 64565200 is out of range [425000000, 1150000000])

Type: Fix
Redmine ID: #501921
Associated modifications:
https://10.10.10.29/c/android/platform/hardware/interfaces/+ /181892
Test: run camera cts

Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Change-Id: I33dca53101350f7680849ae69b0bb603de211b47
```

5.1.2 Camera feature相关

5.1.2.1 测试项

```
android.hardware.camera2.cts.CameraManagerTest#testCameraManagerGetDeviceIdList[1]
```

5.1.2.2 报错日志

```
junit.framework.AssertionFailedError: External camera is not connected on device
with FEATURE_CAMERA_EXTERNAL
    at junit.framework.Assert.fail (Assert.java:50)
    at junit.framework.Assert.assertTrue (Assert.java:20)
    at
    android.hardware.camera2.cts.CameraManagerTest.testCameraManagerGetDeviceIdList (C
ameraManagerTest.java:204)
```

5.1.2.3 解决办法

沟通分析客户反馈没有接USB Camera，但实际跑进去了，说明Camera相关feature中配置支持了USB Camera。

直接先将板上feature文件先删掉试验。

```
rk3576_tablet:/# rm vendor/etc/permissions/android.hardware.camera.external.xml
rk3576_tablet:/# reboot
```

测试pass后，参考修改如下

```
sdk/device/rockchip/rk35xx$ git diff
diff --git a/BoardConfig.mk b/BoardConfig.mk
index 610f2c1..640ecb2 100644
--- a/BoardConfig.mk
+++ b/BoardConfig.mk
@@ -103,9 +103,9 @@ BOARD_RKNN_SUPPORT := false
BOARD_WIDEVINE_OEMCRYPTO_LEVEL := 3
```

```
# camera enable
-BOARD_CAMERA_SUPPORT_EXT := true
+BOARD_CAMERA_SUPPORT_EXT := false
CAMERA_SUPPORT_HDMI ?= false
#CAMERA_SUPPORT_OSD := true
#CAMERA_SUPPORT_VIRTUAL := true
```

5.1.3 时间戳相关

5.1.3.1 测试项

```
android.hardware.camera2.cts.ReadoutTimestampTest#testReadoutTimestamp
```

5.1.3.2 报错日志

```
java.lang.Exception: There were 40 errors:
  java.lang.Throwable(Test failed for camera 125: Camera 125 timestampBase 0
readoutTimestamp (0) should be equal to image timestamp (82754540506000 (expected
= 82754540506000, actual = 0) )
  java.lang.Throwable(Test failed for camera 125: Camera 125 timestampBase 0:
readoutTime (0) - captureStart time (82754540506000) should be > 0)
```

5.1.3.3 解决办法

sdkhardware/rockchip/camera_aidl\$ 目录添加如下修改，如没有更新SDK或找FAE获取

```
commit e57d39c542a7911364ea60ed70cada7cb77a4038
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date:   Fri Aug 23 10:43:40 2024 +0800

    device: usb camera: add readoutTimestamp to fix cts issue

    Test: CtsCameraTestCases

android.hardware.camera2.cts.ReadoutTimestampTest#testReadoutTimestamp

failed log:
readoutTimestamp (0) should be equal to image timestamp
(82754540506000 (expected = 82754540506000, actual = 0) )

Type: Fix
Redmine ID: #501921
Associated modifications:N/A
Test: run camera cts

Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Change-Id: Id6450a0ee7907b9f0940e1c10083ee3500f8912
```

5.1.4 方向相关

5.1.4.1 测试项

```
android.hardware.camera2.cts.ExtendedCameraCharacteristicsTest#testCameraOrientationAlignedWithDevice[1]
```

5.1.4.2 报错日志

```
unit.framework.AssertionFailedError: Camera 100's long dimension must align with screen's long dimension expected:<false> but was:<true>;
    at junit.framework.Assert.fail (Assert.java:50)
    at junit.framework.Assert.failNotEquals (Assert.java:287)
    at junit.framework.Assert.assertEquals (Assert.java:67)
    at junit.framework.Assert.assertEquals (Assert.java:147)
    at
    android.hardware.camera2.cts.ExtendedCameraCharacteristicsTest.testCameraOrientationAlignedWithDevice (ExtendedCameraCharacteristicsTest.java:3309)
```

5.1.4.3 解决办法

修改板上 `/vendor/etc/external_camera_config.xml` 中如下关键字段值

```
<!-- orientation -->
<Orientation degree="90"/> <!-- 0/90/180/270 -->
```

在SDK上的位置: `sdk/device/rockchip/common/external_camera_config.xml`

5.1.5 Android14.0-R5-USB Camera补丁

修复Android14.0 R5版本，USB Camera CTS 通用失败项提交点如下：如果没有找FAE获取

```
提交点1:
commit 8847d6aa76c9445db5094c332dd7d99a7e1240e4
Author: Jian Qiu <qiujian@rock-chips.com>
Date:   Wed Jul 31 15:06:07 2024 +0800

    ExternalCamera: Dealing with V4L2_BUF_FLAG_ERROR

    Try DQBUF again when obtaining V4L2_BUF_FLAG_ERROR

    Type: Fix
    Redmine ID: #497268
    Associated modifications: N/A
    Test: N/A

    Signed-off-by: Jian Qiu <qiujian@rock-chips.com>
    Change-Id: I61d0c0aa69427984b8b2b65841bb9204a57ac83
```

提交点2:

commit d22ac4f020a7499e50ef8e120d77cdcdff837f6a
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date: Wed Aug 7 20:32:46 2024 +0800

device: fix cts stream usage issue

Test: CtsCameraTestCases
android.hardware.camera2.cts.SurfaceViewPreviewTest#testSurfaceSet[1]

Type: Fix
Redmine ID: #500232
Associated modifications: N/A
Test: camera cts

Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Change-Id: Iba303389d5c0d25a6384a421a9b3a20cf198b045

提交点3:

commit 5398546acbb1b814ba0845c8fc7a04c478da6b7a
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date: Mon Aug 12 20:23:40 2024 +0800

device: fix mipi camera cts readoutTimestamp issue

Test: CtsCameraTestCases
android.hardware.camera2.cts.ReadoutTimestampTest#testReadoutTimestamp

Type: Fix/Function
Redmine ID: #500232
Associated modifications: Gerrit links
Test: camera cts

Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Change-Id: I0f3fef0cb576f37fe004006b844adb349987c4df

提交点4:

commit 5186fde7ebc54597b12e98520914597ec18cb7ca
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date: Tue Aug 27 10:26:38 2024 +0800

dev: usb camera: set kBadFramesAfterStreamOn to 4 to fix cts issue

testAllocationFromCameraFlexibleYuv test zero value of frame,
for most of usb camera need skip 2-3 frame to get normal frame;
so use default 4 frame to suite most cameras.

Test: CtsCameraTestCases

android.hardware.camera2.cts.AllocationTest#testAllocationFromCameraFlexibleYuv[1]
]

failed log:

junit.framework.AssertionFailedError: Input allocation data was not
updated
at junit.framework.Assert.fail(Assert.java:50)
at junit.framework.Assert.assertTrue(Assert.java:20)

```
at

android.hardware.camera2.cts.helpers.AssertHelpers.assertArrayNotAllZeroes (Assert
Helpers.java:48)
at

android.hardware.camera2.cts.AllocationTest.validateInputOutputNotZeroes (Allocati
onTest.java:441)
at

android.hardware.camera2.cts.AllocationTest.-$$Nest$mvalidateInputOutputNotZeroes
(UnknownSource:0)
at

android.hardware.camera2.cts.AllocationTest$2$1$1.run (AllocationTest.java:511)
```

Type: Fix
Redmine ID: #501921
Associated modifications: N/A
Test: camera cts

Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Change-Id: Ic93d2e4670bfae601aea525b9fbd841ffa442453

提交点5:

commit 666d5aa4fa154c7454e3d61aa0d09c38a76ad12e
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date: Fri Aug 23 10:21:08 2024 +0800

device: usb camera: default use fpsranges [15:30] fix cts issue

Test: CtsCameraTestCases

```
android.hardware.camera2.cts.CaptureRequestTest#testNoiseReductionModeControl[1]
```

failed log:
Frame duration must be in the range of [500000000, 1000000000],
value 64565200 is out of range [425000000, 1150000000])

Type: Fix
Redmine ID: #501921
Associated modifications:
[https://10.10.10.29/c/android/platform/hardware/interfaces/+/181892](https://10.10.10.29/c/android/platform/hardware/interfaces/+/)
Test: run camera cts

Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Change-Id: I33dca53101350f7680849ae69b0bb603de211b47

提交点6:

commit e0274c8ed40cce941f13d90f811fa013818c2f1f
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date: Fri Aug 23 10:43:40 2024 +0800

device: usb camera: add readoutTimestamp to fix cts issue

Test: CtsCameraTestCases
android.hardware.camera2.cts.ReadoutTimestampTest#testReadoutTimestamp

```
failed log:
readoutTimestamp (0) should be equal to image timestamp
(82754540506000 (expected = 82754540506000, actual = 0) )

Type: Fix
Redmine ID: #501921
Associated modifications:N/A
Test: run camera cts

Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Change-Id: Id6450a0ee7907b9f0940e1c10083ee3500f8912e
```

提交点7:

```
commit 34feb5b65d05a3a8dd860e292d3af17504086dec (tag: android-14.0-mid-rkr6)
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date: Thu Nov 14 16:07:56 2024 +0800
```

device: ExternalCamera: fix usb camera video recording stuck issue

processOneCaptureRequest and FrameWorkerThread::threadLoop not in the same thread, this cause when switch FPS, processOneCaptureRequest do Streamoff, but FrameWorkerThread::threadLoop is still do dqbuf, cause stuck issue

Log:

```
D ExtCamDevSsn: configureV4l2StreamLocked: V4L configuration
format:MJPEG, w 1280, h 720
D ExtCamDevSsn: @v4l2StreamOffLocked: enter!
D ExtCamDevSsn: close mBufFd[0]=22
E ExtCamDevSsn: dequeueV4l2FrameLocked: DQBUF fails: Invalid argument
E ExtCamDevSsn: threadLoop: V4L2 deque frame failed!
D ExtCamDevSsn: close mBufFd[1]=23
D ExtCamDevSsn: close mBufFd[2]=24
E ExtCamDevSsn: dequeueV4l2FrameLocked: DQBUF fails: Invalid argument
```

```
Type: Fix
Redmine ID: # 518321
Associated modifications:
https://10.10.10.29/c/android/rk/hardware/rk35/camera_aidl/+/221884
Test: open camera do video recording.
```

:

```
Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Change-Id: I6a7ba57113202879f8bd2298c24c87073d5de625
```

打补丁位置:

```
sdk/hardware/rockchip/camera_aidl/device$ mm -j12
```

生成/vendor/lib64/camera.device-external-impl-rk.so库，替换板上库后，使用如下命令重启服务生效:

```
adb shell "kill -9 `pidof android.hardware.camera.provider-V1-external-service-rk`"
```

5.2 Android 14.0 MIPI Camera CTS 常见失败项处理办法

5.2.1 camera3_profiles.xml配置错误导致失败项

5.2.1.1 aeAvailableModes配置相关

- 测试项

```
android.hardware.camera2.cts.CaptureRequestTest#testAeModeAndLock[1]
android.hardware.camera2.cts.CaptureRequestTest#testFlashControl[1]
android.hardware.camera2.cts.RobustnessTest#testSimultaneousTriggers[1]
android.hardware.camera2.cts.RobustnessTest#testAeAndAfCausality[1]
android.hardware.camera2.cts.RobustnessTest#testAfThenAeTrigger[1]
android.hardware.camera2.cts.RobustnessTest#testAeThenAfTrigger[1]
android.hardware.camera2.cts.RobustnessTest#testBasicTriggerSequence[1]
```

- 报错日志

```
junit.framework.AssertionFailedError: The static info key
'android.control.aeAvailableModes' values [0] must contain ON mode
    at junit.framework.Assert.fail(Assert.java:50)
    at
    android.hardware.camera2.cts.helpers.StaticMetadata.failKeyCheck(StaticMetadata.java:3051)
    at
    android.hardware.camera2.cts.helpers.StaticMetadata.checkTrueForKey(StaticMetadata.java:3022)
    at
    android.hardware.camera2.cts.helpers.StaticMetadata.getAeAvailableModesChecked(StaticMetadata.java:1326)
    at
    android.hardware.camera2.cts.CaptureRequestTest.testAeModeAndLock(CaptureRequestTest.java:473)
```

- 解决办法

这类报错修改camera3_profiles.xml中关键字,

```
<control.aeAvailableModes value="OFF"/>
```

改成:

```
<control.aeAvailableModes value="ON, OFF"/>
```

5.2.1.2 不带马达摄像头模组测试

- 测试项

```
android.hardware.camera2.cts.CaptureRequestTest#testAfModes[1]
android.hardware.camera2.cts.StillCaptureTest#testFocalLengths[1]
```

- 报错日志

android.hardware.camera2.cts.CaptureRequestTest#testAfModes[1]

```
com.android.ex.camera2.exceptions.TimeoutRuntimeException: Unable to get the
expected result value [2, 6] for key android.control.afState after waiting for
100 results
    at
    android.hardware.camera2.cts.CameraTestUtils.waitForAnyResultValue(CameraTestUtil
s.java:4000)
    at
    android.hardware.camera2.cts.testcases.Camera2SurfaceViewTestCase.waitForAnyResul
tValue(Camera2SurfaceViewTestCase.java:371)
    at
    android.hardware.camera2.cts.CaptureRequestTest.afModeTestByCamera(CaptureRequest
Test.java:2633)
    at
    android.hardware.camera2.cts.CaptureRequestTest.testAfModes(CaptureRequestTest.ja
va:787)
```

android.hardware.camera2.cts.StillCaptureTest#testFocalLengths[1]

```
com.android.ex.camera2.exceptions.TimeoutRuntimeException: Unable to get the
expected result value [0] for key android.lens.state after waiting for 100
results
    at
    android.hardware.camera2.cts.CameraTestUtils.waitForAnyResultValue(CameraTestUtil
s.java:4000)
    at
    android.hardware.camera2.cts.CameraTestUtils.waitForResultValue(CameraTestUtils.j
ava:4023)
    at
    android.hardware.camera2.cts.testcases.Camera2SurfaceViewTestCase.waitForResultVa
lue(Camera2SurfaceViewTestCase.java:348)
    at
    android.hardware.camera2.cts.StillCaptureTest.focalLengthTestByCamera(StillCaptur
eTest.java:759)
    at
    android.hardware.camera2.cts.StillCaptureTest.testFocalLengths(StillCaptureTest.j
ava:716)
```

- 解决办法

如果不支持马达，camera3_profiles.xml中马达相关配置如下：

```
<control.afAvailableModes value="OFF" />
<control.maxRegions value="1,0,0" />
<lens.info.minimumFocusDistance value="0.0" /> <!-- HAL may override this value
from CMC for RAW sensors -->
<request.availableRequestKeys中删除: control.afRegions 字段
<request.availableResultKeys中删除: control.afRegions 字段
```

5.2.1.3 必配参数漏配失败项

- 测试项

```
android.hardware.camera2.cts.CameraDeviceTest#testCameraDeviceVideoSnapshotTemplate[1]
android.hardware.camera2.cts.CameraDeviceTest#testCameraDeviceRecordingTemplate[1]
android.hardware.camera2.cts.CameraDeviceTest#testCameraDevicePreviewTemplate[1]
android.hardware.camera2.cts.CameraDeviceTest#testCameraDeviceStillTemplate[1]
android.hardware.camera2.cts.CameraDeviceTest#testCameraDeviceZSLTemplate[1]
android.hardware.camera2.cts.StaticMetadataTest#testCapabilities[1]
```

- 报错日志

```
java.lang.Exception: Test failed for camera 0: Key android.control.afTrigger
shouldn't be null
    at
    android.hardware.camera2.cts.testcases.Camera2AndroidTestCase.tearDown(Camera2AndroidTestCase.java:169)
    at
    android.hardware.camera2.cts.CameraDeviceTest.tearDown(CameraDeviceTest.java:158)
```

- 解决办法

camera3_profiles.xml中 request.availableResultKeys 和 request.availableResultKeyss 字段，添加关键字 control.afTrigger

5.2.2 拍照帧率相关

5.2.2.1 测试项

```
android.hardware.camera2.cts.CaptureRequestTest#testEdgeModeControl[1]
android.hardware.camera2.cts.CaptureRequestTest#testNoiseReductionModeControl[1]
```

5.2.2.2 报错日志

```
java.lang.Exception: There were 2 errors:
    java.lang.Throwable(Test failed for camera 0: Frame duration must be in the
    range of [33333333, 33333333], value 34311688 is out of range [32833332,
    33833332])
    java.lang.Throwable(Test failed for camera 0: Frame duration must be in the
    range of [33333333, 33333333], value 34334488 is out of range [32833332,
    33833332])
    at
    android.hardware.camera2.cts.testcases.Camera2SurfaceViewTestCase.tearDown(Camera
    2SurfaceViewTestCase.java:176)
    at
    android.hardware.camera2.cts.CaptureRequestTest.tearDown(CaptureRequestTest.java:
    152)
```

5.2.2.3 解决办法

sdkhardware/rockchip/camera\$ 目录参考如下修改,

camera3_profiles.xml 中添加关键字段: sensor.frameDuration

```
commit c8b282f057abb23763f1d8187016e56d3bd0a940
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date: Sat Nov 2 15:48:51 2019 +0800

    camera3_profiles: fix gc2145 following cts failed

    android.hardware.camera2.cts.CaptureRequestTest#testEdgeModeControl

    android.hardware.camera2.cts.CaptureRequestTest#testNoiseReductionModeControl

    Logs:
    Throwable(Test failed for camera 0: Frame duration must be in the range of
    [50000000, 50000000],
    value 61273000 is out of range [49250000, 50750000])

    Change-Id: Ib2ade725e9dbe4d8027998e1017a30c6322b7c28
    Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
```

5.2.3 录像帧率相关

5.2.3.1 测试项

```
android.hardware.camera2.cts.RecordingTest#testBurstVideoSnapshot[1]
android.hardware.camera2.cts.RecordingTest#testIntermediateSurfaceRecording[1]
```

5.2.3.2 报错日志

```
E TestRunner: failed: testBurstVideoSnapshot[1]
(android.hardware.camera2.cts.RecordingTest)
E TestRunner: ----- begin exception -----
E TestRunner: junit.framework.AssertionFailedError: Camera 0: Video frame drop
rate too high: 97.142860%, tolerance 10.000000%. Video size: 1280x720,
expectedDuration [3003.000000,3003.000000], expectedFrameDuration
33.333332, frameDropCnt 34, frameCount 35
E TestRunner: at junit.framework.Assert.fail(Assert.java:50)
E TestRunner: at junit.framework.Assert.assertTrue(Assert.java:20)
E TestRunner: at
android.hardware.camera2.cts.RecordingTest.validateRecording(RecordingTest.java:2
587)
E TestRunner: at
android.hardware.camera2.cts.RecordingTest.validateRecording(RecordingTest.java:2
489)
E TestRunner: at
android.hardware.camera2.cts.RecordingTest.videoSnapshotTestByCamera(RecordingTes
t.java:2062)
```

```

E TestRunner: at
android.hardware.camera2.cts.RecordingTest.videoSnapshotHelper (RecordingTest.java
:1807)
E TestRunner: at
android.hardware.camera2.cts.RecordingTest.testBurstVideoSnapshot (RecordingTest.j
ava:696)
E TestRunner: at java.lang.reflect.Method.invoke (Native Method)
E TestRunner: at
org.junit.runners.model.FrameworkMethod$1.runReflectiveCall (FrameworkMethod.java:
59)

E TestRunner: failed: testIntermediateSurfaceRecording[1]
(android.hardware.camera2.cts.RecordingTest)
W libc : Access denied finding property "sys.hwc.display_pipeline_timeline"
E TestRunner: ----- begin exception -----
E TestRunner: junit.framework.AssertionFailedError: Camera 0: Video duration
doesn't match: recorded 2416.000000ms, expected [1173.333374,1760.000000]ms.
E TestRunner: at junit.framework.Assert.fail (Assert.java:50)
E TestRunner: at junit.framework.Assert.assertTrue (Assert.java:20)
E TestRunner: at
android.hardware.camera2.cts.RecordingTest.validateRecording (RecordingTest.java:2
552)
E TestRunner: at
android.hardware.camera2.cts.RecordingTest.validateRecording (RecordingTest.java:2
489)
E TestRunner: at
android.hardware.camera2.cts.RecordingTest.completeBasicRecording (RecordingTest.j
ava:1681)
E TestRunner: at
android.hardware.camera2.cts.RecordingTest.doBasicRecordingByProfile (RecordingTes
t.java:1615)
E TestRunner: at
android.hardware.camera2.cts.RecordingTest.basicRecordingTestByCamera (RecordingTe
st.java:1581)
E TestRunner: at
android.hardware.camera2.cts.RecordingTest.doBasicRecording (RecordingTest.java:20
1)
E TestRunner: at
android.hardware.camera2.cts.RecordingTest.doBasicRecording (RecordingTest.java:16
6)
E TestRunner: at
android.hardware.camera2.cts.RecordingTest.testIntermediateSurfaceRecording (Recor
dingTest.java:257)

```

5.2.3.3 解决办法

上述帧率失败项，从log看基本都是帧率不够导致。

sdk/hardware/rockchip/camera\$ 目录看下是否有如下修改，没有的话手动打上，然后替换对应库：

```

commit a964c42b8dd34fa918ec215428287ee105ecb201
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date: Fri Nov 15 16:44:43 2024 +0800

    AAL: CameraStream: enable FPS print for debug

```

```

Type: Function
Redmine ID: # N/A
Associated modifications: N/A
Test: logcat | grep FPS

Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Change-Id: I9c89f17f57408bc7a8ed55029d826655b2cd75da
commit a964c42b8dd34fa918ec215428287ee105ecb201
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date:   Fri Nov 15 16:44:43 2024 +0800

AAL: CameraStream: enable FPS print for debug

Type: Function
Redmine ID: # N/A
Associated modifications: N/A
Test: logcat | grep FPS

Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Change-Id: I9c89f17f57408bc7a8ed55029d826655b2cd75da

diff --git a/AAL/CameraStream.cpp b/AAL/CameraStream.cpp
index 3ba2666..411e3db 100644
--- a/AAL/CameraStream.cpp
+++ b/AAL/CameraStream.cpp
@@ -97,11 +97,11 @@ void CameraStream::showDebugFPS(int streamType)
    (double)(1000000000)) / (double)diff;
    switch(streamType) {
        case STREAM_PREVIEW:
-           LOGI("%s: Preview FPS : %.4f: mFrameCount=%d", __func__, fps,
mFrameCount);
+           ALOGI("%s: Preview FPS : %.4f: mFrameCount=%d", __func__, fps,
mFrameCount);
            LOGI("%s, mPendingRequests size=%zu", __func__,
mPendingRequests.size());
            break;
        case STREAM_VIDEO:
-           LOGI("%s: Video FPS : %.4f", __func__, fps);
+           ALOGI("%s: Video FPS : %.4f", __func__, fps);
            break;
        default:
            break;

```

切换到录像模式，点击录像，log查看帧率是否能到30FPS

- 如果到不了，说明摄像头本身数据就有问题，另提redmine排查下驱动和效果文件配置情况。
- 如果能到，将测试机器放置到光照好的地方，不能遮挡摄像头测试。

5.2.4 SurfaceView相关

5.2.4.1 测试项

```
android.hardware.camera2.cts.SurfaceViewPreviewTest#testSurfaceSet[1]
```

5.2.4.2 报错日志

```
Wanted *at least* 30 times:
-> at
android.hardware.camera2.cts.SurfaceViewPreviewTest.verifyCaptureResults(SurfaceViewPreviewTest.java:891)
But was 28 times:
-> at
android.hardware.camera2.impl.CameraCaptureSessionImpl$1.lambda$onCaptureStarted$0(CameraCaptureSessionImpl.java:650)
```

logcat中详细报错日志:

```
E Camera3-OutputStream: getBufferLockedCommon: Stream 13: Can't get next output buffer from buffer manager: Function not implemented (-38)
E Camera3-BufferManager: getBufferForStream: bufferCount (18446744073709551615) exceeds the max allowed buffer count (9) of this stream set
E Camera3-OutputStream: getBufferLockedCommon: Stream 13: Can't get next output buffer from buffer manager: Function not implemented (-38)
E Camera3-BufferManager: getBufferForStream: bufferCount (18446744073709551615) exceeds the max allowed buffer count (9) of this stream set
E Camera3-OutputStream: getBufferLockedCommon: Stream 13: Can't get next output buffer from buffer manager: Function not implemented (-38)
```

5.2.4.3 解决办法

sdkhardware/rockchip/camera_aidl\$ 目录添加如下修改, 如没有更新SDK或找FAE获取

```
commit d22ac4f020a7499e50ef8e120d77cdcdff837f6a
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date:   Wed Aug 7 20:32:46 2024 +0800

    device: fix cts stream usage issue

    Test: CtsCameraTestCases

android.hardware.camera2.cts.SurfaceViewPreviewTest#testSurfaceSet[1]

Type: Fix
Redmine ID: #500232
Associated modifications: N/A
Test: camera cts

Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Change-Id: Iba303389d5c0d25a6384a421a9b3a20cf198b045
```

5.2.5 Camera feature相关

5.2.5.1 测试项

```
android.app.cts.SystemFeaturesTest#testCameraFeatures
```

5.2.5.2 报错日志

```
java.lang.AssertionError. PackageManager#hasSystemFeature should NOT return true  
for android.hardware.camera
```

5.2.5.3 解决办法

从日志分析是板子没有后置摄像头，但配置了后置的feature，需要去掉；参考如下修改：

```
diff --git a/modules/camera_aidl.mk b/modules/camera_aidl.mk  
index 5c18a5be..726b35b8 100644  
--- a/modules/camera_aidl.mk  
+++ b/modules/camera_aidl.mk  
@@ -51,7 +51,6 @@ PRODUCT_COPY_FILES += \  
    endif  
  
    PRODUCT_COPY_FILES += \  
-  
    frameworks/native/data/etc/android.hardware.camera.xml:$(TARGET_COPY_OUT_VENDOR) /  
    etc/permissions/android.hardware.camera.xml \  
  
    frameworks/native/data/etc/android.hardware.camera.front.xml:$(TARGET_COPY_OUT_VE  
NDOR)/etc/permissions/android.hardware.camera.front.xml
```

5.2.6 时间戳相关

5.2.6.1 测试项

```
android.hardware.camera2.cts.ReadoutTimestampTest#testReadoutTimestamp
```

5.2.6.2 报错日志

```
java.lang.Exception: There were 15 errors:
    java.lang.Throwable(Test failed for camera 0: Camera 0timestampBase 0:
readoutTime (249857275125000) - captureStart time (249857272121000) should be
&gt;= exposureTime (4193000))
    java.lang.Throwable(Test failed for camera 0: Camera 0timestampBase 0:
readoutTime (249857312509000) - captureStart time (249857308316000) should be
&gt;= exposureTime (5849000))
    java.lang.Throwable(Test failed for camera 0: Camera 0timestampBase 1:
readoutTime (249857423911000) - captureStart time (249857418062000) should be
&gt;= exposureTime (15998000))
```

5.2.6.3 解决办法

sdkhardware/rockchip/camera_aid1\$ 目录添加如下修改，如没有更新SDK或找FAE获取

```
commit 3ae91a564ad82c41bee71ebf2e65ea24fec853d0
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date:   Mon Aug 12 20:23:40 2024 +0800

    device: fix mipi camera cts readoutTimestamp issue

    Test: CtsCameraTestCases

    android.hardware.camera2.cts.ReadoutTimestampTest#testReadoutTimestamp

    Type: Fix/Function
    Redmine ID: #500232
    Associated modifications: Gerrit links
    Test: camera cts

    Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
    Change-Id: I0f3fef0cb576f37fe004006b844adb349987c4df
```

5.2.7 测试包相关测试项

5.2.7.1 测试项

```
android.hardware.camera2.cts.CameraGPUPrecordingTest#testCameraGpuEncoderPath[1]
```

5.2.7.2 报错日志


```
junit.framework.AssertionFailedError: Camcorder profile should not be null
    at junit.framework.Assert.fail (Assert.java:50)
    at junit.framework.Assert.assertTrue (Assert.java:20)
    at junit.framework.Assert.assertNotNull (Assert.java:218)
    at
    android.hardware.camera2.cts.CameraGPURecordingTest.prepareEncoder (CameraGPURecordingTest.java:174)
    at
    android.hardware.camera2.cts.CameraGPURecordingTest.testCameraGpuEncoderPath (CameraGPURecordingTest.java:106)
```

5.2.7.3 问题分析

经试验media_profiles_V1_0.xml打开720P录像支持后;

如果camera3_profiles.xml中配置后摄则能pass, 配置前置则报错

```
junit.framework.AssertionFailedError: Camcorder profile should not be null
android.hardware.camera2.cts.CameraGPURecordingTest#testCameraGpuEncoderPath[1]
```

问题原因:

CTS测试代码:

```
14_Android_14.0_GMS\cts\tests\camera\src\android\hardware\camera2\cts\CameraGPURecordingTest.java
```

中通过如下获取测试逻辑:

```
    CamcorderProfile profile720p =
    CamcorderProfile.get (CamcorderProfile.QUALITY_720P);
    assertNotNull("Camcorder profile should not be null", profile720p);
```

但实现的地方会判断是否为后摄, 如果不是, 就是NULL;

如果只有一个前置, 就肯定是NULL, 测试逻辑有问题;

```
14_Android_14.0_GMS\frameworks\base\media\java\android\media\CamcorderProfile.java
a
/**
 * Returns the default camcorder profile at the given quality level for the
 * first back-facing
 * camera on the device. If the device has no back-facing camera, this
 * returns null.
 * @param quality the target quality level for the camcorder profile
 * @see #get(int, int)
 * @deprecated Use {@link #getAll} instead
 */
public static CamcorderProfile get(int quality) {
    int numberOfCameras = Camera.getNumberOfCameras();
    CameraInfo cameraInfo = new CameraInfo();
    for (int i = 0; i < numberOfCameras; i++) {
        Camera.getCameraInfo(i, cameraInfo);
        if (cameraInfo.facing == CameraInfo.CAMERA_FACING_BACK) {
            return get(i, quality);
        }
    }
    return null;
}
```

5.2.7.4 解决办法

经确认Android14.0-R4的CTS测试套件上述代码流程有问题，R5-Google更新了测试逻辑就正常了；
对应套件更新提交点

```
From 97c3d42e76661e66832e1d082c9cdafc09ff4a1d Mon Sep 17 00:00:00 2001
From: Bharatt Kukreja <bkukreja@google.com>
Date: Thu, 27 Jul 2023 21:17:10 +0000
Subject: [PATCH] Run test only for cameras which have the given camcorder profile

Currently if the test is run and the camera id does not have the given
camcorder profile (720p), prepareEncoder() fails the assertion to have a
non-null camcorder profile. Make sure to only run the test for camera
ids which have the profile.

Test: atest CameraGPUPrecordingTest.java

Bug: 292216589
Bug: 292216589

Merged-In: I58fe97c64b1e440a9900c93f0366ad9af5dd828d
Change-Id: I58fe97c64b1e440a9900c93f0366ad9af5dd828d
(cherry picked from commit 09c2b077b463b0eaaae4e7c3a2372ab65a73a1b1)
```

5.2.8 权限相关

5.2.8.1 测试项

```
android.permission3.cts.CameraMicIndicatorsPermissionTest#testMicIndicatorWithManualFinishOpStillShows
```

5.2.8.2 报错日志

```
E TestRunner: failed:
testMicIndicatorWithManualFinishOpStillShows (android.permission3.cts.CameraMicIndicatorsPermissionTest)
E TestRunner: ----- begin exception -----
E TestRunner: org.junit.ComparisonFailure: expected:<[Success]> but was:<[Failure [DELETE_FAILED_INTERNAL_ERROR]]>
E TestRunner:    at org.junit.Assert.assertEquals(Assert.java:117)
E TestRunner:    at org.junit.Assert.assertEquals(Assert.java:146)
E TestRunner:    at
android.permission3.cts.CameraMicIndicatorsPermissionTest.uninstall(CameraMicIndicatorsPermissionTest.kt:139)
E TestRunner:    at
android.permission3.cts.CameraMicIndicatorsPermissionTest.tearDown(CameraMicIndicatorsPermissionTest.kt:197)
```

5.2.8.3 解决办法

根据报错信息看下测试逻辑，卸载应用"android.permissionui.cts.apptthataccessescameraandmic"时失败，需使用新PAB包测试；

5.2.9 FOV相关

5.2.9.1 失败项

```
CTS-Verify Camera FOV Calibration
```

5.2.9.2 解决办法

参考如下步骤修改FOV相关参数值。

```
sdk/frameworks/av/services/camera/libcameraservice/apil/client2/Parameters.cpp
```

中调用如下函数计算FOV：

```
res = calculatePictureFovs(&horizFov, &vertFov);
FOV计算公式：
/**
 * Basic field of view formula is:
 * angle of view = 2 * arctangent ( d / 2f )
 * where d is the physical sensor dimension of interest, and f is
 * the focal length. This only applies to rectilinear sensors, for focusing
 * at distances >> f, etc.
 */
```

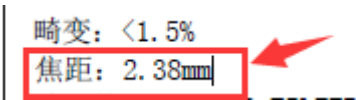
需要修改 camera3_profiles.xml 中的字段：

```
<lens.info.availableFocalLengths value="x.xx"/>
<sensor.info.physicalSize value="x.xx,x.xx"/> <!-- 1600x1.75um 1200x1.75um -->
```

上述两个值获取方法：

- lens.info.availableFocalLengths value

根据模组规格书提供,例如GC2145的一个模组规格书：



- sensor.info.physicalSize value

参考Sensor的Datasheet，例如GC2145， Pixel Size 1.75μm x 1.75μm

Pixel Size	1.75μm x 1.75μm
Active pixel array	1616 x 1232

```
则physicalSize value 中
hori = 1600x1.75um = 2.8 ;
vertical_size = 1200x1.75um =2.1;
<sensor.info.physicalSize value=" 2.8,2.1"/> <!-- 1600x1.75um 1200x1.75um -->
根据FOV计算公式: horizFov = arctan (2.8/2.38) = 49.6度;
注意:
如果实际CTS-Verify测量所需的值为: 56度; 则需要反推修改availableFocalLengths
因为physicalSize value是固定的, 则根据FOV计算公式,
反推出availableFocalLengths值 = 2.8/tan((56°)) = 2.8/1.48 = 1.89;
则可以上述公式, 反推出修改成<lens.info.availableFocalLengths value="1.89"/>即可。
```

5.3 Android 15.0 MIPI Camera CTS 常见失败项处理办法

5.3.1 STS相关测试项

5.3.1.1 测试项

```
StsHostTestCases android.security.sts.Poc20_01#testPocCVE_2019_14104
```

5.3.1.2 报错日志

```
The device has unexpectedly rebooted (41597 seconds after last recorded boot
time, bootreason: kernel_panic,oops:_fatal_exception)
expected to be less than: 1744204232
but was                  : 1744245819
at
com.android.sts.common.tradefed.testtype.SecurityTestCase.tearDown(SecurityTestCa
se.java:160)
```

重启前kernel日志:

```
pc : __pi_strlen+0x124/0x150
lr : strstr+0x34/0x84
sp : fffffffc014e1bb10
x29: fffffffc014e1bb10 x28: 0000000000000000 x27: 0000000000000000
x26: fffffff810c41b688 x25: 0000000000000001 x24: fffffff810c41b515
x23: 00000018000000103 x22: 0000000000000001 x21: 0000000000000006
x20: fffffffc001a13922 x19: 00000018000000103 x18: fffffffc010255060
x17: 00000000fffe8f1f x16: 00000000fffe8f1f x15: 0000007fe705b048
x14: 0000000000000000 x13: 0000000000000000 x12: 0000007fe705b048
x11: 0000000080885658 x10: 00000018000000103 x9 : fffffff81daca4b00
x8 : 0101010101010101 x7 : 7f7f7f7f7f7f7f7f x6 : 746d68736d6e61ff
x5 : 0000000000000006 x4 : 0000000000000003 x3 : 756e69746e6f6300
x2 : 307073696b727f7f x1 : 00000018000000100 x0 : 00000018000000103
```

Call trace:

```
__pi_strlen+0x124/0x150
sdtif_ioctl+0x570/0x810 [video_rkcif]
subdev_do_ioctl_lock+0xcb0/0xf44
video_usercopy+0x44c/0x98c
subdev_ioctl+0x18/0x28
v4l2_ioctl+0x6c/0x84
__arm64_sys_ioctl+0xa8/0xe4
```

```
invoke_syscall+0x58/0x11c
el0_svc_common+0xb4/0xf4
do_el0_svc+0x2c/0xb0
el0_svc+0x2c/0x90
el0t_64_sync_handler+0x68/0xb4
el0t_64_sync+0x1a4/0x1a8
Code: fa4008a0 54ffffde0 17fffffe6 927cec01 (a9400c22)
---[ end trace 0000000000000000 ]---
Kernel panic - not syncing: Oops: Fatal exception
```

5.3.1.3 解决办法

kernel添加如下提交，如没有找FAE获取

```
commit 136c9e13ceae281d48c89a3c1b87f439de3405f2
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date: Mon Jan 13 09:53:56 2025 +0800

media: rockchip: isp: fix android sts CVE_2019_14104 failed issue

fix test android.security.sts.Poc20_01#testPocCVE_2019_14104 cause
kernel panic issue

The issue was reported in https://redmine.rock-chips.com/issues/531294
The kernel crash log:
[ T7364] Unable to handle kernel paging request at virtual address
0000001800000100
...
pc : __pi_strlen+0x124/0x150
lr : strstr+0x34/0x84
...

Call trace:
__pi_strlen+0x124/0x150
sditf_ioctl+0x570/0x810 [video_rkcif]
subdev_do_ioctl_lock+0xcb0/0xf44
video_usercopy+0x44c/0x98c
subdev_ioctl+0x18/0x28
v4l2_ioctl+0x6c/0x84
__arm64_sys_ioctl+0xa8/0xe4
invoke_syscall+0x58/0x11c
el0_svc_common+0xb4/0xf4
do_el0_svc+0x2c/0xb0
el0_svc+0x2c/0x90
el0t_64_sync_handler+0x68/0xb4
el0t_64_sync+0x1a4/0x1a8
Code: fa4008a0 54ffffde0 17fffffe6 927cec01 (a9400c22)
---[ end trace 0000000000000000 ]---

Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Change-Id: I26f2da26d9aec7b2f1dfa93c1165d7579b89f3ba
```

5.3.2 CtsVirtualDevicesCameraTestCases 相关

5.3.2.1 测试项

```
android.virtualdevice.cts.camera.VirtualCameraNdkTest#getCameraIds_virtualDevice
android.virtualdevice.cts.camera.VirtualCameraNdkTest#availabilityCallbacks_virtualDevice
```

5.3.2.2 报错日志

```
<Module name="CtsVirtualDevicesCameraTestCases" abi="arm64-v8a" runtime="8063"
done="true" pass="0" total_tests="2">
  <TestCase name="android.virtualdevice.cts.camera.VirtualCameraNdkTest">
    <Test result="fail" name="getCameraIds_virtualDevice">
      <Failure message="missing (1): 1">
        <StackTrace>missing (1): 1
---
expected   : [1]
but was    : []
at
  android.virtualdevice.cts.camera.VirtualCameraNdkTest.getCameraIds_virtualDevice(
VirtualCameraNdkTest.java:148)</StackTrace>
      </Failure>
    </Test>
    <Test result="fail" name="availabilityCallbacks_virtualDevice">
      <Failure message="Wanted but not invoked:">
        <StackTrace>Wanted but not invoked:
mMockAvailabilityCallback.onCameraAvailable(
  "1"
);
- &gt; at
  android.virtualdevice.cts.camera.VirtualCameraNdkTest.availabilityCallbacks_virtualDevice(VirtualCameraNdkTest.java:164)
Actually, there were zero interactions with this mock.

at
  android.virtualdevice.cts.camera.VirtualCameraNdkTest.availabilityCallbacks_virtualDevice(VirtualCameraNdkTest.java:164)</StackTrace>
      </Failure>
    </Test>
  </TestCase>
</Module>
```

5.3.2.3 问题分析

测试代码位置:

```
sdk\cts\tests\tests\virtualdevice\camera\src\android\virtualdevice\cts\camera\VirtualCameraNdkTest.java中getCameraIds_virtualDevice
```

测试失败问题原因:

```
List<String> ndkCameras = Arrays.asList(nativeCameraManager.getCameraIds());
```

获取到的List<String> ndkCameras size为0, 导致
assertThat(ndkCameras).containsExactly(FRONT_CAMERA_ID);失败;

nativeCameraManager.getCameraIds()调用流程为:

```

----->
Sdk\meworks\av\camera\ndk\impl\ACameraManager.cpp CameraManagerGlobal::getCameraIdList(
-----
CameraManagerGlobal::onStatusChangedLocked()
-----》
CameraManagerGlobal::supportsCamera2ApiLocked()
-----
sdk\frameworks\av\services\camera\libcameraservice\CameraService.cpp-
CameraService::supportsCameraApi
----- CameraService::getDeviceVersion()-----》
sdk\frameworks\av\services\camera\libcameraservice\common\CameraProviderManager.cpp
----- CameraProviderManager::getHighestSupportedVersion()---
去匹配CameraID;异常, 从而失败;
CameraID匹配异常原因:
是virtual_camera注册的cameraID为: v0/1/1_1000+deviceId值;
注册代码逻辑在:
sdk\frameworks\av\services\camera\virtualcamera\VirtualCameraService.cc中
createCameraId(const int32_t deviceId)
注册上去的CameraID名称为:
kCameraIdPrefix + std::to_string(deviceId) + "_" +
std::to_string(sNextIdNumericalPortion++);
其中:
kCameraIdPrefix[] = "v";
static std::atomic_int sNextIdNumericalPortion{1000};
deviceId是根据虚拟设备的ID;

而下发匹配的CameraID为:
Sdk\frameworks\av\camera\ndk\impl\ACameraManager.cpp
中mCameraService->addListener(mCameraServiceListener, &cameraStatuses);里面
hardware::CameraStatus的cameraId, 打印获取到的为0/1

综合分析:
CtsVirtualDevicesCameraTestCases测试代码为Google 原生代码;
虚拟摄像头的实现逻辑: 在 sdk/frameworks/av/services/camera/virtualcamera$ framework
层, vendor改不了;

总结: CtsVirtualDevicesCameraTestCases为Android15.0 新增测试项, 测试失败为测试包本身测试
逻辑问题;
平板能测试过, 是因为刚好平板有前后摄像头注册的cameraID String为: 0/1刚好匹配上了.

```

5.3.2.4 解决办法

目前这两项失败项属于豁免。

6. ITS

6.1 ITS各测试项官方说明

[ITS各测试项官方说明](#)

6.2 ITS 测试报告查看

- 示例报告

单个测试场景示例报告汇总文件:

CameraITS_4_luokzb\cam_id_0\scene2_a中文件scene_test_summary.txt

```
Cam0 scene2_a
SKIP test_auto_flash.py
SKIP test_display_p3.py
SKIP test_effects.py
PASS test_format_combos.py
PASS test_jpeg_quality.py
SKIP test_num_faces.py
FAIL test_preview_min_frame_rate.py
SKIP test_reprocess_uv_swap.py
```

- 报告解读

```
Cam0 scene2_a
//Camera ID 0, 测试scene2_a 场景
SKIP/PASS/FAIL
```

6.3 ITS常见failed项处理办法

6.3.1 帧率问题

- 测试项

```
scene2_a test_preview_min_frame_rate.py
```

- 报错日志

```
Traceback (most recent call last):
  File "/home/fih/USER/ITS_14/lib/python3.9/site-packages/mobly/base_test.py",
line 756, in exec_one_test
    test_method()
  File "/home/fih/USER/ITS_14/android-cts-verifier-14_r4-linux_x86-arm/android-
cts-verifier/CameraITS/tests/scene2_a/test_preview_min_frame_rate.py", line 133,
in test_video_min_frame_rate
    raise AssertionError('\n'.join(errors))
AssertionError: Preview frame rate was 30.001912873639593, expected to be 15,
ATOL: 2.
```

- 解决办法

调整json效果文件中帧率动态范围，支持从15-30动态波动。